

Betaxanthin pathway – LaTeX rendering preview

2026.08.02

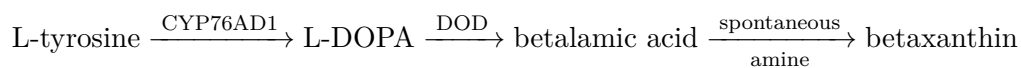
Source note: notes/scratch.2026.08.02.215928-dopa.md. Every variant below uses only `amsmath`, already loaded by `paper/nature-biotech/preamble.tex` (line 25).

A. Minimal – plain arrow chain



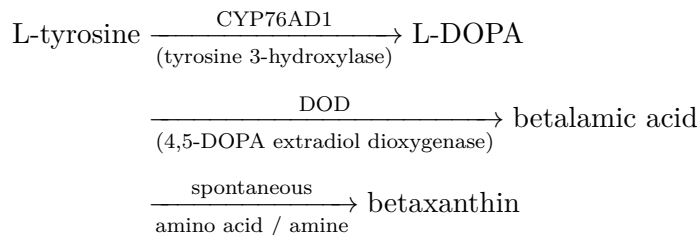
```
\[
  \text{L-tyrosine} \longrightarrow \text{L-DOPA} \longrightarrow
  \text{betalamic acid} \longrightarrow \text{betaxanthin}
\]
```

B. Enzyme-annotated, one line



```
\[
  \text{L-tyrosine}
  \xrightarrow{\text{CYP76AD1}}
  \text{L-DOPA}
  \xrightarrow{\text{DOD}}
  \text{betalamic acid}
  \xrightarrow[\text{amine}]{\text{spontaneous}}
  \text{betaxanthin}
\]
```

C. Full annotation, wrapped (fits an 88 mm column)

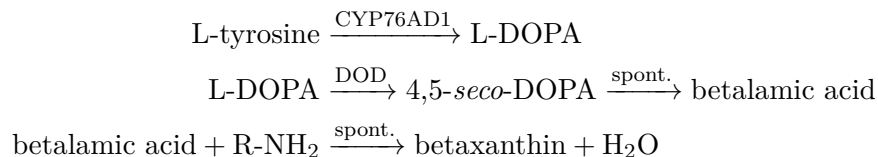


```

\begin{equation*}
\begin{aligned}
&\text{\text{L-tyrosine}} \\
&\quad \&\xrightarrow[\text{(tyrosine 3-hydroxylase)}]{\text{CYP76AD1}} \\
&\quad \text{\text{L-DOPA}} \quad \ll[4pt] \\
&\quad \&\xrightarrow[\text{(4,5-DOPA extradiol dioxygenase)}]{\text{DOD}} \\
&\quad \text{\text{betalamic acid}} \quad \ll[4pt] \\
&\quad \&\xrightarrow[\text{amino acid / amine}]{\text{spontaneous}} \\
&\quad \text{\text{betaxanthin}}
\end{aligned}
\end{equation*}

```

D. Mechanistic – intermediate and co-substrate explicit

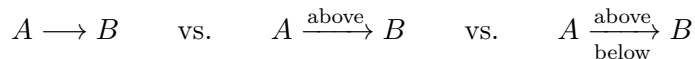


```

\begin{equation*}
\begin{aligned}
&\text{\text{L-tyrosine}} \\
&\quad \&\xrightarrow{\text{CYP76AD1}} \text{\text{L-DOPA}} \quad \ll \\
&\text{\text{L-DOPA}} \\
&\quad \&\xrightarrow{\text{DOD}} \text{\text{4,5-}\textit{seco}\text{-DOPA}} \\
&\quad \&\xrightarrow{\text{spont.}} \text{\text{betalamic acid}} \quad \ll \\
&\text{\text{betalamic acid}} + \text{\mathrm{R}\text{-}\mathrm{NH}_2} \\
&\quad \&\xrightarrow{\text{spont.}} \text{\text{betaxanthin}} + \text{\mathrm{H}_2\mathrm{O}}
\end{aligned}
\end{equation*}

```

Side-by-side of the two arrow commands



`\longrightarrow` is a fixed-length arrow; `\xrightarrow` stretches to fit its labels, which is why the enzyme-annotated versions space out unevenly. That is expected and correct – each arrow is exactly as wide as its own label.